FACULDADE DE ENGENHARIA DA UNIVERSIDADE DO PORTO

Design of an Asynchronous Flash Analog-to-Digital Converter

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INTRODUCTION TO RESEARCH

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Resumo

Mudar para português o abstract

Abstract

The proliferation of internet of things and biomedical sensors has created a high demand for low-power signal processing interfaces. In many of these applications, the signals of interest are sparse, characterized by long periods of inactivity. Classical synchronous analog to digital converters are inherently inefficient in such scenarios, as they consume dynamic power continuously due to the global clock, regardless of the input signal activity. Ideally the sampling rate should dynamically be adapted to the signal properties, so that power consumption due to the clock activity could be minimized. However, an adaptive sampling rate is not straightforward. As such, the use of an asynchronous ADC is a natural approach to sample non-uniform signals. Asynchronous ADCs require architectural changes, to allow proper operations without a global clock. The adoption of asynchronous control in Flash architectures is primarily motivated by the need to eliminate power consumption high-speed clock distribution networks, thereby improving energy efficiency. Furthermore, asynchronous operation offers intrinsic benefits in handling comparator metastability and reducing latency, making it particularly attractive for high-speed, event-driven applications where signal activity varies significantly over time. This dissertation proposes the design and implementation of an asynchronous Flash ADC. By removing the clock, the proposed architecture aligns power consumption with the input signal activity, theoretically achieving a much lower power consuption.

To address this, an offline trimming strategy is proposed to calibrate the comparator offsets without compromising the high-speed operation of the flash topology. The work encompasses the theoretical analysis, schematic design, and validation of the system through analog/mixed-signal co-simulation. The expected outcome is a robust, clockless ADC architecture that offers a superior figure of merit for sparse signal applications in compared to conventional synchronous architectures.

Keywords: Analog-to-Digital Converter, Asynchronous Design, Flash ADC, Level-Crossing Sampling, Offset Trimming, Low Power, IoT.

UN Sustainable Development Goals

The United Nations Sustainable Development Goals (SDGs) provide a global framework to achieve a better and more sustainable future for all. It includes 17 goals to address the world's most pressing challenges, including poverty, inequality, climate change, environmental degradation, peace, and justice.

Electronic systems play a pivotal role in modern society, acting as the backbone for smart infrastructure, healthcare monitoring, and environmental sensing. However, the massive deployment of battery-operated devices poses a significant challenge regarding energy consumption and electronic waste. This dissertation contributes directly to the efficiency and sustainability of these systems.

The specific Sustainable Development Goals addressed by this work are:

SDG 7 Affordable and Clean Energy: Ensure access to affordable, reliable, sustainable and modern energy for all. By optimizing the power consumption of data converters, we extend the battery life of devices and reduce the overall energy footprint of IoT networks.

SDG 9 Industry, Innovation and Infrastructure: Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation. This work advances the state-of-the-art in microelectronics by proposing novel asynchronous architectures that enable smarter and more efficient industrial sensors.

SDG	**Target**	**Contribution**	**Performance Indicators and Metrics**
7	7.3	By double the global rate of improvement in energy efficiency, this work reduces the power waste in standby modes of electronic sensors.	Reduction in Energy per Conversion (fJ/conv) compared to synchronous architectures.
9	9.5	Enhance scientific research, upgrade the technological capabilities of industrial sectors, encouraging innovation.	Successful validation of the proposed calibration algorithm and circuit topology.

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"Our greatest glory is not in never falling, but in rising every time we fall"

Confucius

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List of Acronyms

IoT	Internet of Things
ADC	Analog-to-Digital Converter
DAC	Digital-to-Analog Converter
CMOS	Complementary Metal-Oxide-Semiconductor
LSB	Least Significant Bit
MSB	Most Significant Bit
SNDR	Signal-to-Noise-and-Distortion Ratio
SNR	Signal-to-Noise Ratio
SQNR	Signal-to-Quantization-Noise Ratio
ENOB	Effective Number of Bits
DNL	Differential Non-Linearity
INL	Integral Non-Linearity
BER	Bit Error Rate
FoM	Figure of Merit
SAR	Successive Approximation Register
LCS	Level-Crossing Sampling
GS/s	Giga-Samples per Second
MS/s	Mega-Samples per Second
SHA	Sample-and-Hold Amplifier
V_{th}	Threshold Voltage
V_{OS}	Input Offset Voltage
V_{DD}	Supply Voltage
INC	Increment event signal (upper level crossing)
DEC	Decrement event signal (lower level crossing)
DOUT	Digital output code
DOUT+	Upper digital threshold code
DOUT-	Lower digital threshold code
LC	Level Crossing
LFP	Local Field Potential
AP	Action Potential
RMSE	Root-Mean-Square Error

Chapter 1

Introduction

This chapter provides a brief introduction to the topic associated with the work carried out during the dissertation period. A contextualization is given, followed by a presentation of the research question and the main motivations for carrying out this study. The main objectives of this dissertation will also be outlined, as well as the methodology used to achieve them. Finally, the structure of the dissertation is presented.

We live in an era where smart devices are everywhere, known as the Internet of Things, IoT. While the processing and storage of information are inherently digital, benefiting from the scaling of Moore's Law, the physical world remains fundamentally analog. Physical variables such as temperature, sound and biological potentials are continuous in both time and amplitude. Consequently, the Analog-to-Digital Converter, ADC, serves as the critical interface bridging these two domains, enabling digital systems to interact with the real world [65].

Energy efficiency has shifted from being a secondary performance metric to a primary design constraint, many of the sensors operate on limited battery or rely on energy harvesting, where every Joule of power dissipation has a direct effect on the system's autonomy. In this landscape, the data conversion block almost always takes place as the most spender in this field, especially in sensor interfaces where the digital transmission is duty-cycled.

However, a fundamental characteristic of many environmental and physiological signals is sparsity in between pulses. Signals such as electrocardiograms, voice activity, or environmental monitoring data are spread in bursts, they contain short periods of high information content followed by long intervals of inactivity.

Traditional data conversion approaches, based on uniform sampling and dictated by the Nyquist-Shannon theorem [113], treat these silence periods with the same computational power and energetic waste as the active periods. This creates a paradigm of inefficiency, the system dissipates power to generate redundant digital samples that carry no new information. Recognizing and exploiting this sparsity is the key to unlocking ultra-low-power electronic interfaces.

Figure 1.1: Time-sparse signal representation and sampling strategies. (a) Time-sparse signal example showing burst activity with periods of inactivity. (b) Continuous-time event-driven sampling and conversion, which responds to signal changes. (c) Conventional uniform sampling operating at fixed Nyquist rate regardless of signal activity. (d) Discrete-time sampling with event-driven conversion, combining periodic and event-based approaches. This illustration demonstrates the efficiency of adaptive sampling methods for sparse signals compared to uniform sampling at the Nyquist rate [81].

The time-sparse nature of many signals can be observed in the burst waveform where information is concentrated in short intervals separated by long idle periods, as was demonstrated in 1.1. In particular, subfigures (b) and (d) highlight how continuous-time and discrete-time event-driven strategies avoid generating samples during inactivity, in contrast with conventional uniform sampling which keeps operating at a fixed Nyquist rate regardless of the signal content.

1.1 Motivation and Problem Statement

The energy waste in massive-scale IoT deployments and the inefficiency of conventional synchronous architectures when processing sparse data shows the need for a fundamental shift toward architectures that activate only when meaningful data events occur.

1.1.1 Inefficiency of Synchronous Sampling

Traditional Analog-to-Digital Converters operate under a fixed-rate sampling rate, dictated by a global clock signal (f_s). As established by the Nyquist-Shannon sampling theorem [113], this rate is determined by the maximum possible bandwidth of the signal, $f_s \geq 2B$. However, in many real-world applications, the signal is often active for only a fraction of the time, meaning the true information content is much lower than the theoretical maximum bandwidth suggests.

This leads to a massive waste of energy due to the fundamental relationship governing dynamic power consumption in CMOS circuits [9]:

$$P_{dynamic} = \alpha \cdot f \cdot C_L \cdot V_{DD}^2 \tag{1.1}$$

where:

- $P_{dynamic}$ is the dynamic power dissipated.
- α is the activity factor (or switching factor).

- f is the clock frequency.
- C_L is the load capacitance being switched (e.g., in the clock tree).

In a uniformly clocked ADC, f is fixed by the Nyquist criterion and the clock network toggles every cycle, so α remains high even when the input signal is almost static. As a result, the converter continues to charge and discharge large clock and comparator capacitances at every period, dissipating dynamic power to generate redundant samples that do not carry new information. For example, if a sensor signal is active only 10% of the time but the ADC samples continuously, 90% of the switching events are spent tracking intervals where the input is essentially constant, directly illustrating how synchronous operation leads to massive energy waste for non synchronous signals.

1.2 Objectives and Scope of the Dissertation

Based on the limitations identified in traditional synchronous data conversion when dealing with asynchronous sensor signals, the primary objective of this work is to design and implement an energy-efficient ADC architecture, so this ADC must exploit signal inactivity to achieve a significant reduction in power consumption.

To achieve the overall goal of developing an efficient ADC, the following specific objectives are defined for this dissertation:

1. Asynchronous Architecture Design: To develop the circuit-level design of an asynchronous Flash ADC, minimizing dynamic power consumption by ensuring that power is only dissipated when an input signal event occurs.
2. Offline Trimming Implementation: To implement an offline trimming mechanism capable of detecting and compensating for the input offset voltage of the comparators.
3. Mixed-Signal Validation: To validate the complete system through analog and digital co-simulation. This validation must confirm the functionality and accuracy of both the asynchronous core and the trimming logic.
4. Performance Benchmarking: To quantify the energy efficiency of the proposed design using well known figures of merit. The results must be compared against the current state of the art synchronous ADCs and relevant asynchronous solutions.

In order to support these objectives, the state-of-the-art review is structured to progressively narrow the focus from general data-conversion theory to the specific asynchronous Flash architecture and trimming techniques addressed in this work. Chapter 2 begins by revisiting the fundamentals of sampling, quantization, and performance metrics for data converters, establishing the theoretical background required to interpret figures of merit such as SNDR, ENOB, and Walden FoM. It then surveys the main synchronous ADC architectures (Flash, SAR, Pipeline, and Sigma-Delta), highlighting their inherent power, speed and resolution trade-offs and identifying why conventional clocked Flash converters become inefficient when dealing with sparse signals.

1.3 Document Structure

Chapter 2

Literature Review

Subsequent sections of the literature review focus on asynchronous and level-crossing sampling strategies, examining prior work on event-driven converters and their advantages for time-sparse signals, and finally discuss existing calibration and trimming techniques for comparator offset reduction, providing the reference framework against which the proposed bulk-driven resistive trimming mechanism will be compared.

This chapter presents the state-of-the-art in Analog-to-Digital Converters, reviewing theoretical foundations and analyzing survey data to justify the proposed architecture.

2.1 Fundamentals of Analog-to-Digital Conversion

This section provides the theoretical background required to understand the operation and performance characterization of data converters. It covers the fundamental steps of the conversion process such as sampling, quantization, and coding.

2.1.1 The Data Conversion Process

2.1.1.1 Ideal Data Conversion

The analog-to-digital conversion process acts as the bridge between the continuous physical world and the discrete digital domain. Conceptually, it involves two distinct operations: discretization in time, sampling, and discretization in amplitude, quantization. Ideally, this process should be instantaneous and lossless within the signal bandwidth of interest [65].

2.1.1.2 The Sampling Operation

Sampling converts a continuous-time signal $x(t)$ into a discrete-time sequence $x[n]$ [65].

According to the Nyquist-Shannon sampling theorem [113], a band-limited signal with maximum frequency B can be perfectly reconstructed if the sampling frequency f_s satisfies:

$$f_s \geq 2B \tag{2.1}$$

Violating this condition results in aliasing, where high-frequency spectral components mixes into the baseband, becoming indistinguishable from the original signal.

2.1.1.3 The Quantization Operation

While sampling transforms continuous time in discrete values, quantization maps the continuous amplitude of each sample to one of a finite number of levels [65]. An N-bit ADC divides the input range (V_{ref}) into 2^N discrete levels. The step size between adjacent levels is the Least Significant Bit (LSB):

$$V_{LSB} = \frac{V_{ref}}{2^N} \tag{2.2}$$

Unlike sampling, quantization is non-reversible and introduces a deterministic error. This error is typically modeled as white noise added, the quantization noise with a uniform probability distribution. For an ideal quantizer, the Signal-to-Quantization-Noise Ratio (SQNR) is given by the formula:

$$SQNR_{dB} \approx 6.02N + 1.76 \tag{2.3}$$

2.1.1.4 Coding

The final stage is encoding the quantized level into a binary format. The choice of coding scheme being the most common straight binary and gray code which depends on the system requirements for data processing and transmission, but does not affect the fundamental analog performance [9].

2.1.2 Classical Performance Metrics

To evaluate and compare different data converters objectively, a standard set of metrics is used. These are categorized into dynamic and static parameters.

2.1.2.1 Resolution and Sampling Rate

Resolution defines the theoretical dynamic range, while the sampling rate determines the maximum signal bandwidth. However, these are nominal values the actual performance is limited by noise and non-linearities [65].

2.1.2.2 Signal-to-Noise-Distortion Ratio (SNDR)

Signal to noise ratio adding distortion ratio or SNDR is the primary dynamic metric. It is the ratio of the signal power to the total power of all noise and harmonic distortion components. From SNDR, the Effective Number of Bits, ENOB, is derived [65]:

$$ENOB = \frac{SNDR_{dB} - 1.76}{6.02} \tag{2.4}$$

This value represents the true resolution of the converter at a specific input frequency.

2.1.2.3 Differential and Integral Non-Linearity (DNL and INL)

Static linearity is characterized by measuring the deviation of code transition levels from their ideal positions [65]. The DNL measures the deviation of a single step width from the ideal 1 LSB. A DNL less than -1 LSB implies a missing code in the transfer function.

The INL is the cumulative sum of DNL errors, representing the deviation of the transfer curve from a straight line. Specific INL patterns like saw-tooth shapes can reveal systematic errors in the architecture, such as gain mismatch or non-linear biasing [101].

2.1.2.4 Offset and Gain Error

These are linear errors, offset is a constant shift of the transfer characteristic as seen before in section 1.2.3, while gain error is a deviation in the slope [65]. Unlike non-linearity, these errors preserve the signal shape and can often be calibrated out simply. Linear imperfections such as offset and gain error modify the global position and slope of the transfer characteristic without changing its shape, as shown in Fig. 2.1. In this representation, a constant DC shift moves the whole transfer curve up or down (offset error), while a slope deviation alters the full-scale span, the so called gain error, both of which can be compensated by simple digital calibration.

Figure 2.1: Adapted from [94] ADC transfer function showing offset and gain errors. Offset error represents a constant DC shift applied uniformly across all codes, while gain error modifies the slope of the transfer function. Both are linear errors that can be calibrated using digital post-processing techniques [65].

2.1.2.5 Figures of Merit

Figures of merit (FoMs) provide normalized metrics to compare ADC performance across different architectures and technologies. The Walden FoM quantifies energy efficiency per conversion step:

$$\text{FoM}_W = \frac{\text{Power}}{2^{\text{ENOB}} \cdot f_s} \tag{2.5}$$

where lower values indicate better performance.

The Schreier FoM extends this by incorporating bandwidth effectiveness for oversampled converters:

$$\text{FoM}_S = \text{SNDR} + 10\log_{10}\left(\frac{\text{BW}}{\text{Power}}\right) \tag{2.6}$$

with higher values representing superior performance.

2.1.3 State-of-the-Art Comparative Analysis

The analytical plots and tables presented below incorporate data from an exhaustive set of 140 state-of-the-art works. All entries were selected from the ISSCC and JSSC publications, due to their high prestige and stringent peer-review process in the field of data converters. Within these sources, only papers reporting both supply voltage (V_{DD}) and signal-to-noise-and-distortion ratio (SNDR) were retained, ensuring that all points in the performance envelopes are based on consistently documented electrical and dynamic performance metrics. To ensure transparency and reproducibility, the complete list of references used to generate these performance envelopes, covering Flash, SAR, Pipeline, and Sigma-Delta architectures, is provided in [1–8, 10–14, 16–21, 23–28, 30–40, 42–60, 62, 63, 66, 67, 69–80, 82–91, 93, 95, 96, 98, 100, 102–105, 109–112, 114–119, 121–123, 125–139, 142–161].

The architectures marked with an [1] are Flash ADCs representing the best-case performance for their respective plots among all surveyed works. These highlight the performance envelope limits achieved by Flash converters in each figure. They serve as benchmarks for comparing other architectures like SAR, Pipeline, and Sigma-Delta.

The evolution of the Walden Figure of Merit as a function of sampling rate for different ADC architectures is summarized in Fig. 2.2. This plot allows identifying the performance envelope across a wide range of f_s and highlights how specific implementations approach the energy-efficiency limit at their target bandwidth.

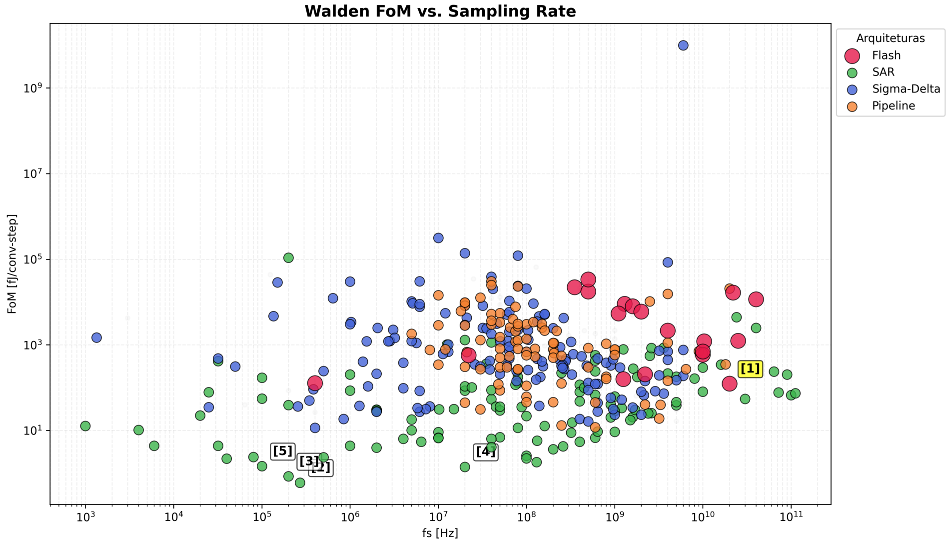

Figure 2.2: Walden FoM vs. Sampling Frequency (Flash reference [1] highlighted)

Table 2.1: Walden FoM vs. Sampling Frequency (Flash reference [1] highlighted)

Ref.	Architecture	Publication	f_s [Hz]
[1]	**Flash**	**Lukas [92]**	**20G**
[2]	SAR	Jang [61]	270k
[3]	SAR	Ginsburg [41]	200k
[4]	SAR	Cano [29]	20M
[5]	SAR	Muller [97]	200

2.1.4 Metrics Adaptation for Asynchronous Systems

2.2 Sampling Paradigms: Synchronous vs. Asynchronous

2.2.0.1 Non-Uniform Sampling and Event-Driven Systems

2.2.0.2 Uniform Sampling and Nyquist-Rate Systems

2.2.0.3 Signal Sparsity and Theoretical Energy Efficiency Benefits

2.3 Synchronous Architectures

To address asynchronous ADCs we first need to address synchronous ADCs, they have a global clock signal that dictates sampling instances and synchronizes internal operations. While these architectures are highly mature and widely used, they face fundamental power efficiency trade-offs, particularly when operating at high frequencies or processing sparse data [65].

2.3.1 Synchronous Flash ADC

The Flash ADC is conceptually the simplest and fastest architecture, performing a complete conversion in a single clock cycle [9]. It utilizes a resistive ladder to generate $2^N - 1$ reference voltages, which are compared simultaneously to the input signal by a large bank of comparators. This parallel comparison produces a thermometer code, that a digital logic block then converts into a standard binary output. Its primary limitation is the exponential increase in hardware complexity, area, and power consumption as resolution increases.

The thermometer term refers to the parallel output from the Flash ADC comparators, where '1's run up to the input voltage level like mercury in a thermometer, indicating signal strength before being converted to standard binary by an encoder.

2.3.1.1 The Power Bottleneck

The primary drawback of the synchronous Flash architecture is its exponential scaling. Survey data clearly illustrates that high speed translates into high power in these designs. For example, a 6-bit 500 MS/s CMOS Flash ADC reported in 1999 already consumed 400 mW [120].

More extreme cases, such as a 22 GS/s 5-bit design, can consume up to 3W [108].

The continuous clocking of a massive comparator bank creates a significant energy dissipation even when the input signal is static. This lack of adaptivity makes synchronous Flash architectures increasingly unsuitable for battery-powered or energy-harvesting applications.

2.3.2 SAR (Successive Approximation Register)

The SAR ADC operates using a binary search algorithm [101], for each sample the internal logic tests one bit at a time, starting from the most significant bit (MSB). A single comparator compares the input to the output of an internal digital to analog converter (DAC), if the input is higher, the bit is set to '1' otherwise, it is set to '0'. This process repeats for N cycles. Because it uses very few active components, it is the most energy efficient choice for low speeds.

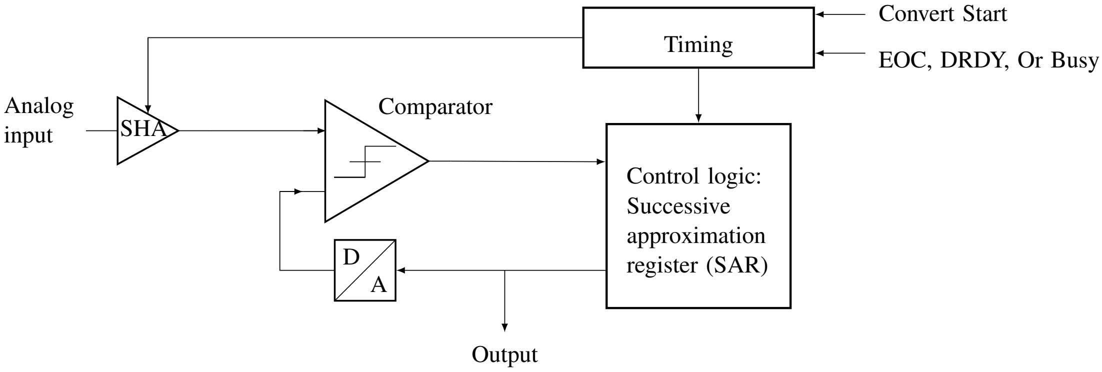

Figure 2.3: Successive Approximation Register (SAR) ADC architecture showing the sample-and-hold amplifier (SHA), comparator, digital-to-analog converter (DAC), and control logic. The SAR logic iteratively approximates the input signal by testing each bit from MSB to LSB, achieving high energy efficiency at moderate speeds [101].

The binary search operation of a SAR ADC is implemented by successively testing each bit through the interaction of the sample-and-hold amplifier (SHA), DAC, comparator, and control logic, as shown in Fig. 2.3. At every conversion step, the SAR logic updates the DAC output and the comparator decides whether the input is above or below the trial voltage, gradually converging from the MSB to the LSB towards the final digital code.

2.3.3 Pipeline

The Pipeline ADC breaks the conversion into several sequential stages [65], each stage resolves a few bits of information, quantizes them, and passes the remaining error, the residue, to the next stage after amplification. This approach allows the ADC to work on multiple samples concurrently, obtaining very high throughput and high resolution, the problem relies in the inherent processing latency.

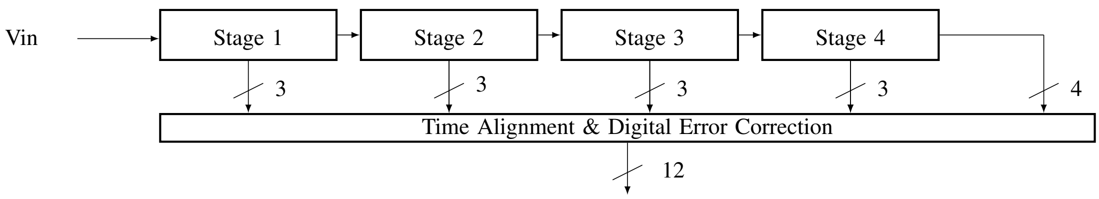

Figure 2.4: Pipeline ADC architecture showing multiple conversion stages operating in parallel on different input samples. Each stage resolves a subset of bits, generates a partial digital output, and amplifies the residue for the next stage. The pipelined structure enables high-throughput operation at the expense of increased latency and circuit complexity.

The pipelined converter processes multiple samples simultaneously by distributing the resolution across several cascaded stages, as shown in Fig. 2.4. Each stage resolves a subset of bits, generates a partial digital output, and amplifies the residue for the next stage, while a digital error-correction block aligns the timing of all stage outputs to produce the final high-throughput conversion result.

2.3.4 Sigma-Delta (ΣΔ)

The ΣΔ architecture relies on oversampling, sampling much faster than the Nyquist rate [15], and noise-modulation. A modulator integrates the difference between the input signal and a feedback version of the quantized output. This process pushes the quantization noise into higher frequencies, outside the signal band of interest, a digital filter then removes the high-frequency noise, resulting in high resolution and precision [107].

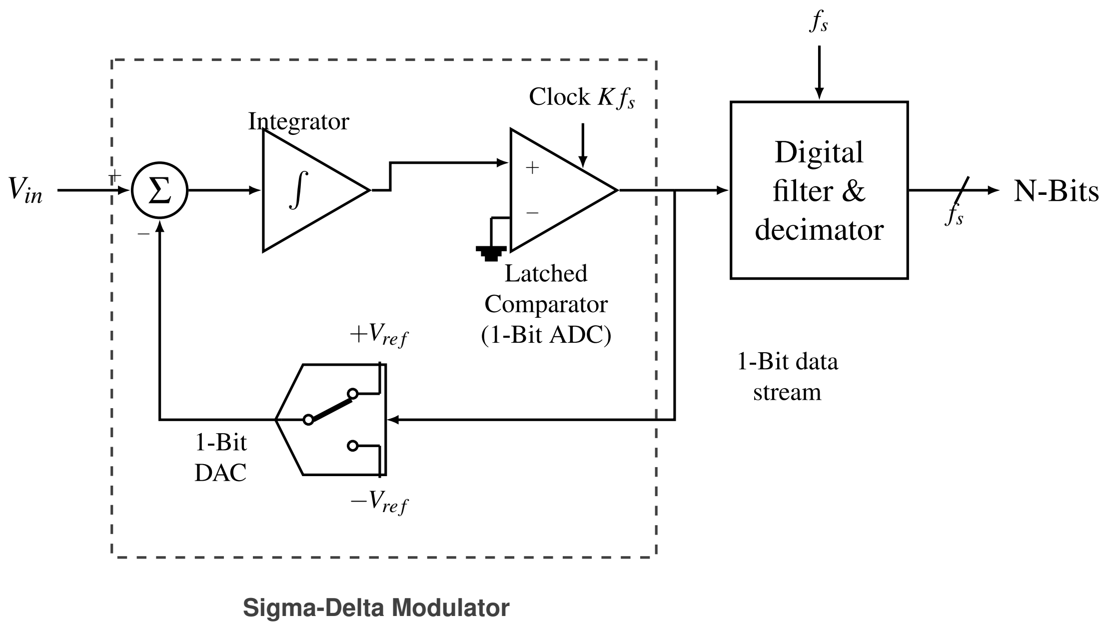

Figure 2.5: First-order sigma-delta ADC architecture illustrating the oversampling modulator with summer, integrator, 1-bit quantizer, feedback DAC, and digital decimation filter, which together implement noise shaping and high-resolution conversion [15, 107].

Oversampling and noise shaping are achieved by the feedback loop formed by summer, integrator, 1-bit quantizer, and digital decimation filter in the first-order sigma-delta architecture, as shown in Fig. 2.5. The modulator continuously integrates the difference between the analog input and the 1-bit DAC feedback, pushing most of the quantization noise to high frequencies, which are later removed by the digital filter.

2.3.5 Dual-slope

This integrating architecture [106] measures the time required for a capacitor to charge and discharge, in the first phase, the capacitor is charged by the input voltage for a fixed period then for the second phase, it is discharged by a known reference voltage. The time it takes to return to zero is proportional to the average value of the input signal it's highly accurate and immune to high-frequency noise but is much slower than other architectures.

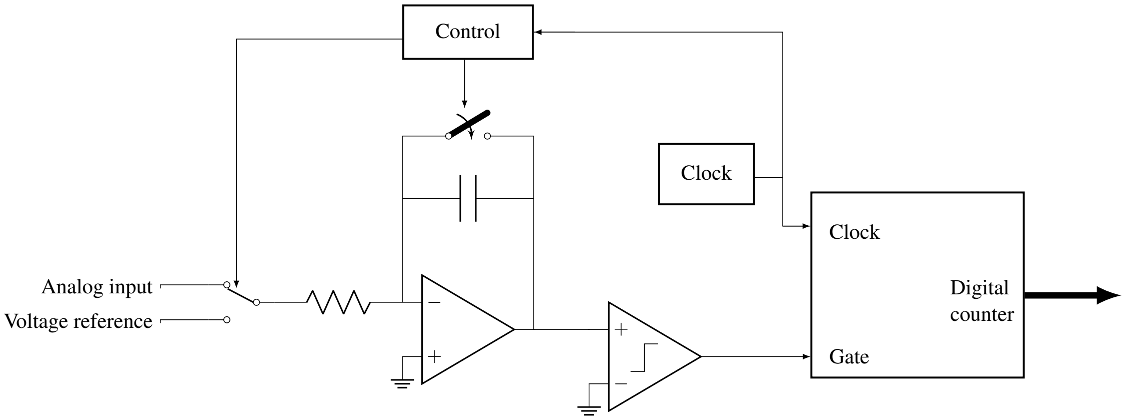

Figure 2.6: Dual-slope (integrating) ADC architecture showing the charging phase where an unknown input signal is integrated for a fixed time, and the discharge phase where a precise reference voltage drives the capacitor back to zero. The conversion time is proportional to the input voltage, providing excellent accuracy and noise rejection at the expense of low throughput.

The dual-slope architecture converts the input voltage into time by integrating it during a fixed interval and then discharging the capacitor with a known reference, as shown in Fig. 2.6. The digital counter measures the discharge interval controlled by the reference voltage, so the resulting count is proportional to the average value of the input signal and inherently rejects high-frequency noise.

2.4 Asynchronous Architectures

Asynchronous, or event-driven, architectures represent a fundamental paradigm shift in data conversion [81], unlike synchronous ADCs, which are bound by a global clock and the Nyquist-Shannon sampling theorem, asynchronous ADCs operate based on the signal's activity. This approach offers a more efficient alternative for processing sparse signals.

2.4.1 Level-Crossing Sampling (LCS)

The core principle behind many asynchronous ADCs is Level-Crossing Sampling (LCS) [81]. In traditional uniform sampling, the signal is captured at fixed time intervals (T_s), and the amplitude is quantized. In LCS, the process is inverted: the amplitude levels are fixed (quantization thresholds), and the ADC records the exact time at which the input signal crosses these thresholds.

This method is particularly powerful for signals that remain constant or change slowly over long periods. Instead of generating redundant samples that capture no new information, the LCS ADC remains idle, only producing a digital event when the signal effectively changes by more than the defined threshold.

Figure 2.7: Digital logic core of a level-crossing asynchronous ADC showing the two DACs that generate upper and lower thresholds around the input signal V_{IN}, the pair of comparators that produce the INC and DEC event signals, and the digital logic that updates the output code $DOUT$ and the time-stamp Δt based on each level crossing. The feedback paths from the digital logic to the DACs keep the thresholds centered around the most recent output level, enabling event-driven tracking of the input signal instead of uniform-rate sampling [99].

The core of the proposed event-driven conversion scheme is illustrated in Fig. 2.7. In this architecture, two on-chip DACs generate a pair of voltage thresholds placed symmetrically around the analog input signal V_{IN}, while two comparators continuously monitor the crossings of V_{IN} with respect to these levels. The upper comparator asserts the INC signal whenever V_{IN} rises above the upper threshold, whereas the lower comparator asserts the DEC signal when V_{IN} falls below the lower threshold.

Upon each level-crossing event, the digital logic block updates the encoded amplitude word $DOUT$ and the control words driving the DACs so that the thresholds are recentered around the new signal level. In addition, the logic measures the elapsed time since the previous event and outputs a time-stamp Δt, thereby representing the signal as a sequence of non-uniform events $(DOUT, \Delta t)$ instead of uniformly spaced samples. This mechanism concentrates conversion activity in intervals where the input varies rapidly and naturally suppresses conversions during flat regions, enabling an asynchronous ADC operation whose power consumption scales with the actual signal dynamics rather than with a fixed global sampling clock.

2.4.2 Flash ADC

The Asynchronous Flash ADC adapts the parallel structure of a standard Flash ADC but removes the sampling clock entirely. In this architecture, the comparators are not latched by a clock, they operate in continuous time, constantly monitoring the input voltage V_{in} against the reference ladder. The event generation happens when the input signal crosses a threshold, the corresponding comparator changes its output state immediately, this transition triggers an asynchronous logic to generate a digital output pulse.

2.5 Clockless Event Generation

2.5.1 The Voltage-to-Current Conversion Concept in Event Detection

2.5.2 Current Comparator Operating Principles

2.5.3 Topologies for Event Detection in the Literature

2.5.3.1 Current-Mirror-Based Current Comparator

The current-mirror-based comparator represents the simplest class of current-mode comparators. In this architecture, the input current and a reference current are applied to two branches of a mirror network that generates a difference current, which is then converted into a voltage across a high-resistance node. This voltage variation is sensed by a subsequent amplifier or inverter, producing a logic output that indicates whether the input is larger or smaller than the reference. The small-signal behaviour is governed by the output resistance of the mirror devices: a high intrinsic resistance allows small current differences to produce significant voltage variations, which improves sensitivity.

The main advantage of this structure is its simplicity, very small silicon area and low design effort, since it relies on basic current mirrors and standard logic inverters. It can also offer good static power characteristics if the bias currents are kept small, and the absence of complex feedback paths simplifies stability considerations. However, the high output resistance that improves sensitivity also leads to poor frequency response when driving capacitive loads, so the addition of inverters or buffers to obtain rail-to-rail output significantly degrades speed. Their propagation delay can vary strongly with process and temperature, making them usually less suitable for very high-speed asynchronous Flash ADCs unless carefully optimised.

2.5.3.2 Traff Current Comparator

The Traff comparator introduces a high-speed continuous-time current comparison scheme based on a source-follower input stage and a CMOS inverter with positive feedback. The input current is first converted into a voltage at a low-impedance node through a pair of source followers, this node then drives an inverting stage that regenerates the decision. The positive feedback loop keeps the input node from slewing rail-to-rail, so its voltage excursion is limited around a quiescent point and the effective input impedance remains low, which shortens the time required to develop the voltage difference that triggers the output transition. When the differential input current exceeds the resolution threshold, the latch-like action of the feedback rapidly pulls the output to one of the rails, providing a logic-compatible decision.

Among its main advantages, this architecture achieves a high speed, with reduced propagation delay for a given input current step and relatively low power consumption for high-speed operation. Furthermore, the absence of large high-resistance nodes improves the frequency response and makes the design attractive for high-bandwidth current-mode ADCs. On the other hand, the Traff comparator exhibits a deadband region for very small input currents in which both input transistors operate near cut-off, temporarily increasing input resistance and degrading response time making it not suitable for this application.

2.5.3.3 Lu Chen Current Comparator

The continuous-time comparator proposed by Lu Chen employs a CMOS complementary amplifier with resistive feedback, followed by two resistive-load amplifiers and CMOS inverters. The core stage uses a pair of complementary transistors operating in saturation, with a MOS transistor in the linear region acting as a feedback resistor between input and output nodes. Small-signal analysis of this structure shows that, for typical device sizing, both input and output resistances of the complementary amplifier are approximately inversely proportional to the sum of the transconductances of the input devices, which strongly reduces the voltage swing at the internal nodes during a comparison. The following resistive-load amplifiers provide additional gain with relatively low static current, while the final inverters restore rail-to-rail logic levels.

This design achieves a favourable speed–power ratio because the reduced internal voltage swings translate into shorter response time for a given input current, while the resistive-load stages limit power consumption, particularly when the input current increases. Compared with other high-speed comparators, simulations reported by the authors show that the average delay for small currents is comparable to more complex solutions, but with lower or similar power, and without requiring external bias currents or voltages, improving process robustness. As a drawback, the achievable resolution remains dependent on the sizing of the feedback device and the effective transconductance of the input pair, so aggressive scaling of bias currents to save power can degrade sensitivity.

2.5.3.4 Hysteresis Current Comparator

The hysteresis current comparator studied by Ranjana Sridhar introduces a current gain stage with non-linear feedback and a CMOS inverter output. The gain stage consists of cascaded resistive amplifiers and an inverter enclosed by a pair of transistors that implement non-linear feedback, whose gates are driven by the output node while their sources sense the difference-node voltage. For small $|I_{\text{diff}}|$, both feedback transistors are off and the node behaves as a capacitive load around a quiescent operating point, whereas for larger positive or negative differences one of the feedback devices turns on, enhancing the loop gain around that point and accelerating the transition.

This non-linear feedback mechanism creates an effective hysteresis, once the output has switched, a finite current difference of opposite polarity is required to bring the internal node back to the decision boundary, which improves noise immunity and metastability behaviour. The use of multiple gain stages and feedback devices increases circuit complexity and area compared with simpler comparators, and the hysteretic behaviour must be carefully dimensioned so that the added memory effect does not introduce excessive offset or distort the transfer characteristic in applications where strictly monotonic behaviour is required.

2.5.3.5 Min and Kim Current Comparator

The Min and Kim comparator employs a resistive feedback network around a current-source inverting amplifier to reduce input and output impedances and improve dynamic response. The structure consists of several current-source amplifiers connected in cascade, with the first stage incorporating a resistor that feeds back a portion of the output voltage to the input node. In steady state, this feedback lowers the effective impedance seen at the input and output, thereby limiting the voltage swing required to represent a given current difference and shortening the time needed to reach a decision threshold at the amplifier output. A final CMOS inverter converts the amplified voltage into a digital logic level.

Simulation results reported in the literature indicate that this architecture achieves low propagation delay and good process immunity, with the resistive feedback helping to stabilise the operating point and reduce sensitivity to parameter variations. Additionally, the reduced node swings make the comparator suitable for relatively high-speed applications as long as the bias currents are chosen appropriately. A key limitation, however, is that both speed and resolution are strongly dependent on the bias current of the current-source amplifiers: achieving very high resolution requires higher bias currents, which directly increases static power dissipation and degrades energy efficiency. Moreover, because the design relies on current-source loads rather than resistive loads, its power consumption does not decrease when the input current increases, unlike other comparators where dynamic behaviour can partially trade off with static consumption.

2.5.3.6 CC-II-Based Current Comparator

The CC-II-based comparator proposed by Veepsa Bhatia uses a current subtractor followed by a current conveyor of the second generation and a multi-inverter output stage. The current subtractor is implemented with a p-type current mirror that produces the difference between the input current and a reference current across a resistor, effectively converting the difference into a proportional voltage. This voltage is then applied to the low-impedance X-terminal of a CC-II, whose behaviour is characterised by a unity voltage gain from Y to X and current conveyance from X to Z, so that the current difference is sensed at X and reproduced at the high-impedance Z terminal. The inverter chain at the output converts the conveyor's analog voltage into full-swing logic levels suitable for digital processing.

A key feature of this architecture is the use of positive feedback from one of the inverters back to the Y terminal of the conveyor, so that small variations in the X-node voltage induced by the input current difference are reinforced through the CC-II action. This positive feedback shortens the response time and increases sensitivity to small current steps, allowing the comparator to achieve sub-5 μA resolution with nanosecond-level propagation delay in simulations. The number of devices is also moderate, and the CC-II realisation provides inherently low input impedance at X and high output impedance at Z. On the downside, the presence of the op-amp or amplifier block used to realise the CC-II, together with the feedback network, increases design complexity and power compared with very simple mirror-based comparators. Furthermore, the performance is sensitive to the bandwidth and stability of the conveyor implementation, which must be carefully compensated to prevent oscillations due to the positive feedback.

2.6 Calibration and Trimming Techniques

In high-speed Flash ADCs, the accuracy of the system is fundamentally limited by the precision of the comparators. While the architecture fundamentals were established in the previous sections, practical implementations must address the non-idealities of the fabrication process, specifically the Input Offset Voltage (V_{os}).

2.6.1 The Component Mismatch Problem

In deep sub-micron CMOS technologies, transistors that are drawn with identical dimensions on the layout will exhibit slight differences in their electrical parameters after fabrication. This phenomenon, known as mismatch, affects the threshold voltage (V_{th}) and the current gain factor (β) of the differential pair in a comparator [9].

According to Pelgrom's Law cited in section 2.3.3 of [65], the standard deviation of the threshold voltage mismatch (σ_{Vth}) is inversely proportional to the square root of the transistor area ($W \cdot L$):

$$\sigma_{V_{th}} = \frac{A_{V_{th}}}{\sqrt{W \cdot L}} \tag{2.7}$$

Where $A_{V_{th}}$ is a technology-dependent constant. This creates a critical trade-off, to minimize offset without calibration, transistors must be made large, which increases parasitic capacitance and degrades the ADC speed and power efficiency. Therefore, small transistors are used for speed, and calibration is needed to correct the resulting offset.

2.6.2 Calibration Classifications

Calibration techniques can be broadly categorized by their timing and domain [9]. In terms of timing there can be derived two types foreground calibration which interrupts normal operation to measure and correct errors and background calibration which operates continuously but adds significant complexity. In terms of domain there is digital calibration that corrects the output code mathematically, whereas analog calibration adjusts the circuit biasing or load conditions to nullify the offset at the source.

2.6.3 Trimming Techniques

2.6.3.1 Internal Loading

The internal resistive loading method involves placing a variable resistive network in parallel or in series with the output loads of the comparator stage. By digitally switching small resistors or MOS switches operating in the triode region in parallel with the load branch, the effective resistance R_L is modulated. If the ADC has an offset, the trimming network is adjusted to introduce an equal and opposite value to effectively zero the error. This technique is typically implemented using a binary-weighted bank of PMOS transistors as the variable resistance, where the digital control code is determined at startup and stored in a register.

2.6.3.2 Reference Ladder Trimming

The reference ladder trimming technique corrects comparator offset by utilizing the main reference ladder as a calibration source [141]. In this architecture, a switching network selects specific voltages from the resistor ladder and applies them to the bulk terminals of the input transistors M1 and M2. This approach leverages the body effect to shift the threshold voltage (V_{th}) of the differential pair, thereby nullifying the input-referred offset caused by process mismatch.

As demonstrated in the simulation results of the threshold voltage ($|V_t|$) versus the ladder tap voltage (LT), the threshold voltage of transistors M1 and M2 increases linearly from approximately 356.5 mV to 372 mV as the calibration voltage is adjusted. In contrast, the threshold voltage of transistors M3 and M4 remains constant at approximately 356.5 mV because their bulk terminals are not connected to the tuning network. This targeted adjustment of the bulk potential for the input differential pair provides a tuning range sufficient to compensate for random process variations and ensure the accuracy of the comparator trip points.

Figure 2.8: Bulk-driven resistive reference ladder trimming architecture used for comparator offset calibration. The preamplifier input pair (M1, M2) has its bulk terminals driven by a selectable tap of the main reference resistor ladder via the offset calibration circuit (OSCAL). A digital control word selects one of the ladder taps (Rtap) through a tree decoder and multiplexers, generating a calibration voltage V_{cal} that modulates the bulk node V_B. By shifting the bulk potential of M1 and M2 while keeping the load devices at a fixed substrate potential, the threshold voltage of the input pair is adjusted through the body effect, providing a tunable range to cancel process-induced input offset in high-speed Flash ADC comparators [140].

The effectiveness of the bulk-driven trimming mechanism is illustrated by the controlled change of the input pair threshold voltage with the ladder tap voltage, as shown in Fig. 2.9 and Fig. 2.8. As the calibration voltage LT increases, the threshold of M1 and M2 shifts linearly while M3 and M4 remain fixed, providing the tuning range required to compensate comparator offset caused by random process variations [140].

2.6.4 Advantages of Resistive Trimming for Asynchronous ADCs

Compared to dynamic offset-cancellation techniques such as auto-zeroing, which rely on accurately timed clock phases ϕ_1 ϕ_2 and additional storage capacitors that must be periodically charged and discharged [9], resistive trimming is particularly well suited to an event-driven asynchronous architecture. Once the calibration bits are determined during a dedicated foreground calibration phase, the resistive trimming network remains static, meaning that no further switching activity or

Figure 2.9: Threshold voltage V_{th} variation of input differential pair transistors M1, M2 and load transistors M3, M4 as a function of the resistive ladder tap voltage LT. The graph demonstrates the body effect as the bulk potential of M1 and M2 is modulated via the reference ladder, providing a calibration range to compensate for process-induced mismatch [140].

clocking is required and the comparator bias conditions are held purely by DC conduction paths. This static behavior ensures that the trimming circuitry does not disturb the intrinsic event-driven operation of the ADC, which should ideally remain idle whenever the input signal is not crossing a quantization threshold.

Another important advantage is that the resistive network can be connected so as to avoid placing large additional capacitances on the high-speed nodes of the comparator, thereby preserving the original small-signal bandwidth and regeneration speed. In contrast, dynamic calibration schemes typically introduce extra sampling capacitors and switch parasitics at sensitive nodes, which can degrade the comparator's response time and limit the maximum achievable sampling rate in high-speed Flash architectures [9].

2.7 Asynchronous vs. Synchronous

2.7.1 Advantage

The primary motivation for adopting asynchronous architectures is the direct relationship between signal activity and power consumption [81].

In a synchronous system, the clock tree and the comparators switch at every clock cycle, regardless of whether the input is changing, the power consumption is dominated by the fixed clock frequency [9]:

$$P_{sync} \approx f_{clk} \cdot C_{total} \cdot V_{DD}^2 \tag{2.8}$$

In an asynchronous ADC, the switching frequency is replaced by the event rate. If the signal is constant or slowly varying, the comparators and digital logic remain static, leading to the following proportionality:

$$P_{async} \propto Activity \times V_{DD}^2 \tag{2.9}$$

This property ensures that the energy consumed is always proportional to the information content of the signal which makes these ADCs significantly more efficient for sparse signal environments, where they can achieve near-zero power during periods of inactivity.

2.7.2 Comparative Evaluation

An analysis of state-of-the-art converters shows that asynchronous ADCs excel in energy efficiency across a wide range of sampling rates, particularly for low-to-medium bandwidths [81].

While Synchronous Flash ADCs are designed for peak performance at a specific frequency and suffer from a much higher power consumption caused by the continuous clocking, asynchronous designs scale their power consumption linearly with the input activity.

Table 2.2: Comparison of Power Consumption between Synchronous and Asynchronous Flash ADCs. Simulation results from this work showing the power advantage of asynchronous operation for sparse signals.

Topology	**Average power (P [W])**
Synchronous	0.1016
Asynchronous	0.0314

The impact of this architectural difference on long-term technology trends is illustrated in Figure 2.10. The plot depicts the energy per conversion (P/f_s) over the last two decades. While both paradigms show a downward trend due to process node scaling, asynchronous designs (indicated by triangular markers) consistently occupy the lower bound of the energy envelope. This suggests that eliminating the global clock is a key enabler for breaking the power efficiency barriers in modern designs.

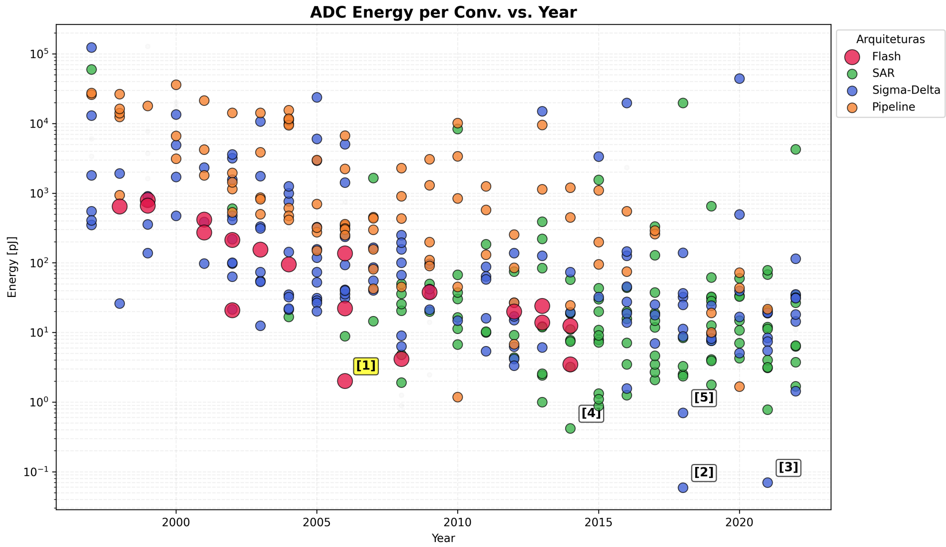

Figure 2.10: Energy per Conversion vs. Year (Flash reference [1] highlighted)

The long-term reduction in energy per conversion step driven by architectural innovation and technology scaling is depicted in Fig. 2.10. Each point in the plot corresponds to a published design, making it possible to compare how different ADC architectures fill the energy year plane and to identify the most efficient solutions. From the previous analysis, it becomes clear that

Table 2.3: Energy per Conversion vs. Year (Flash reference [1] highlighted)

Ref.	Architecture	Publication	Energy [pJ]
[1]	**Flash**	**Chen [22]**	**2.02**
[2]	Sigma-Delta	Jiang [64]	0.059
[3]	Sigma-Delta	Veldhoven [124]	0.070
[4]	SAR	Ginsburg [41]	0.420
[5]	Sigma-Delta	Kim [68]	0.703

synchronous architectures face a fundamental inefficiency when dealing with time-sparse sensor signals: the global clock forces continuous switching activity, so dynamic power remains proportional to the clock frequency even during long periods with no new information content. In contrast, event-driven schemes naturally decouple power consumption from the Nyquist rate and tie it instead to the actual rate of relevant signal transitions. Combining this observation with the survey data on state-of-the-art converters, where Flash ADCs define the speed frontier but suffer from poor Walden FoM at high sampling rates, while more energy-efficient SAR and sigma-delta architectures are optimized for lower bandwidths, motivates the choice of an asynchronous Flash topology as the core of this work. By removing the global sampling clock and operating comparators in a level-crossing, event-driven manner, the proposed asynchronous Flash ADC aims to preserve the intrinsic low-latency and high-speed advantages of the Flash architecture, while leveraging signal sparsity to significantly improve energy efficiency and close the FoM gap with the best-performing synchronous designs in the targeted operating region.

2.7.3 Chapter Summary and Design Directives

Asynchronous Flash ADCs combine a parallel comparison front-end with event-driven operation to exploit the time-sparse nature of many sensor signals. In these architectures, the input is processed as a current and the conversion is triggered only when the signal crosses predefined decision levels, avoiding unnecessary activity during idle periods. Compared with conventional synchronous converters, this enables a significant reduction in dynamic power, since large capacitive nodes associated with sampling clocks and comparator arrays are only switched when relevant information is present in the input waveform. In addition, operating in current mode relaxes voltage headroom constraints, facilitates low-voltage design and can offer improved speed for a given technology and power budget, making this approach well suited for energy-constrained IoT interfaces.

From an architectural point of view, the proposed converter follows a Flash topology, where a bank of comparators observes different reference levels in parallel and produces a thermometer-like code that is subsequently encoded into binary. As in a conventional Flash ADC, this guarantees single-cycle conversion and very low latency, but here the global clock is removed and the decision activity is instead driven by input events. This choice transfers the power bottleneck from the sampling network to the design of the current comparators themselves, since their intrinsic delay, input range and power consumption directly determine the achievable resolution, maximum event rate and overall energy efficiency. For this reason, a detailed study and comparison of several current comparator architectures was carried out before selecting the most suitable structure for the asynchronous core.

Chapter 3

Proposed Asynchronous Flash ADC Design

3.1 Architectural Evolution and Methodology

The design of the voltage comparator responsible for generating the digital output code of the analog-to-digital converter was not arrived at directly. It evolved through a sequence of candidate topologies, each evaluated against the constraints imposed by the target supply voltage, the required conversion accuracy, the effective input dynamic range, and the overall demands on circuit compactness. The following sections document this progression, presenting each topology that was considered and explaining the physical reasoning that either led to its adoption or prompted its replacement by a subsequent candidate.

3.2 Phase I: The Baseline Current-Mirror OTA Flash Configuration

3.2.1 Operating Principle and Core Topology

3.2.2 Current Comparator Topology Exploration and Characterization

The first topology investigated for the comparator core was the current-mirror flash ADC (CMF-ADC) introduced by Lee *et al.* [**lee2017**], originally proposed as the sensing front end of a coarse-fine dual-loop digital low-dropout regulator. Although the application context of that work differs substantially from the present design, the operating principle of the CMF-ADC constitutes a direct voltage-to-thermometer-code conversion mechanism that is directly applicable in the context of multi-bit comparison, making it a natural starting point for the architectural exploration.

The circuit is divided into two cascaded functional stages: a transconductance cell and a current-to-code converter. The transconductance cell is implemented as a differential pair biased by a tail current of $2I_B$. Its function is to translate the voltage difference between the input voltage, V_{in}, and an adjustable reference voltage, $V_{\text{ref,c}}$, into an error current I_{err}, yielding branch currents $I_1 = I_B - I_{\text{err}}$ and $I_2 = I_B + I_{\text{err}}$, where a positive I_{err} corresponds to V_{in} exceeding $V_{\text{ref,c}}$. A third current, I_3, replicates I_1 through a current mirror and is subsequently distributed to the conversion stage.

The current-to-code converter generates a multi-bit thermometer output by replicating I_2 and I_3 at each comparison node through PMOS and NMOS current mirrors, respectively, sized with deliberately asymmetric multiplication factors. In the five-bit implementation described in [**lee2017**], the PMOS mirror ratios decrease monotonically from $\times 14$ to $\times 6$ across the five output nodes specifically $\times 14$, $\times 12$, $\times 10$, $\times 8$, and $\times 6$ for bits $C_{\mathrm{LPT}}[4]$ through $C_{\mathrm{LPT}}[0]$ while the NMOS mirror ratios increase correspondingly from $\times 6$ to $\times 14$. At each comparison node, the output logic level is determined by whichever mirrored current dominates: when the PMOS contribution exceeds the NMOS contribution, the node is driven to a logic high; otherwise, the NMOS pull-down prevails and the output settles to a logic low.

To explain the operating principle clearly, it is most instructive to consider a configuration with an odd number of output bits, of which the five-bit case is the natural choice. An odd bit count provides a well-defined, symmetric midpoint whose comparison threshold aligns exactly with the condition $V_{\mathrm{in}} = V_{\mathrm{ref,c}}$, making the midpoint of the thermometer code unambiguous and facilitating a direct account of the circuit behaviour at the reference voltage. When V_{in} lies substantially below $V_{\mathrm{ref,c}}$, the error current I_{err} is strongly negative, meaning $I_2 < I_B$ while $I_3 > I_B$. Since the NMOS mirrors replicate I_3 with larger multiplication factors than the PMOS mirrors replicate I_2 at every node, the NMOS current dominates throughout and the output code resolves to 00000.

As V_{in} rises toward $V_{\mathrm{ref,c}}$, the magnitude of I_{err} decreases and the comparison nodes with the highest PMOS-to-NMOS mirror ratio are the first to transition. The node $C_{\mathrm{LPT}}[4]$, carrying a PMOS mirror of ratio $\times 14$ against an NMOS mirror of only $\times 6$, presents the most favourable balance in favour of the PMOS side and therefore requires only a modest positive shift in I_2 relative to I_3 before its output transitions high. Node $C_{\mathrm{LPT}}[3]$ ($\times 12$ PMOS, $\times 8$ NMOS) follows at a moderately larger error current, while the least significant nodes, $C_{\mathrm{LPT}}[1]$ and $C_{\mathrm{LPT}}[0]$, carry progressively more dominant NMOS mirrors and require a correspondingly larger current excess before their outputs transition.

At the exact condition $V_{\mathrm{in}} = V_{\mathrm{ref,c}}$, the error current vanishes and both I_2 and I_3 are equal to I_B. Nodes $C_{\mathrm{LPT}}[4]$ and $C_{\mathrm{LPT}}[3]$ deliver PMOS currents of $14I_B$ and $12I_B$, respectively, against NMOS contributions of $6I_B$ and $8I_B$, so both outputs are high. Nodes $C_{\mathrm{LPT}}[1]$ and $C_{\mathrm{LPT}}[0]$ exhibit the reverse relationship and remain low. The central node $C_{\mathrm{LPT}}[2]$ carries equal PMOS and NMOS currents of $10I_B$ and is therefore in an exactly balanced. The resulting thermometer code at $V_{\mathrm{in}} = V_{\mathrm{ref,c}}$ is consequently 11?00, where the question mark denotes the indeterminate state of the central bit, fully consistent with the expected behaviour of a comparator operating precisely at its decision threshold.

The thresholds governing the transition of each bit are fully determined by the mirror ratios, as established analytically in [**lee2017**], and result in a monotonic, symmetric thermometer-coded response across the input voltage range. This architecture combines inherently fast conversion speed stemming from the absence of a conventional error amplifier, whose finite bandwidth would otherwise limit the response time of the control loop [**lee2017**] with a straightforward, hardware-efficient structure and an explicitly quantifiable input-output characteristic. It was therefore adopted as the first candidate for the present comparator design, and its operating principle informed all subsequent topological choices.

3.3 Phase II: The Complementary Dual-Input Flash Configuration

3.3.1 Motivation for Wide Input Common-Mode Range Extension

3.3.2 Voltage Headroom Limitations and Performance Degradation in Low-Voltage Nodes

A notable limitation of the CMF-ADC topology, as configured, is that its input common-mode range is restricted by the headroom requirements of the differential pair and tail current source. In order to extend the effective operating range of the comparator toward full rail-to-rail input coverage, a complementary dual-input architecture was developed and evaluated as a second candidate.

This topology employs two parallel transconductance stages operating simultaneously: one based on an NMOS differential input pair and a second based on a PMOS differential input pair. The motivation for this arrangement is well established in the context of rail-to-rail input amplifiers [**razavi**]: an NMOS-input stage operates correctly when the common-mode input voltage is in the vicinity of the negative supply rail, where the gate overdrive of the input transistors is sufficient to maintain saturation, while a PMOS-input stage provides coverage near the positive supply rail. Operated jointly, the two stages guarantee that at least one transconductor remains fully functional across the complete supply voltage span.

In the implementation explored here, the two transconductors were not combined through a simple parallel superposition of their output currents. Rather, one output branch of each OTA was directed toward the PMOS current mirrors responsible for generating the individual bits of the thermometer code, following the same current-comparison principle established in the CMF-ADC. The remaining output branch of each OTA was cross-coupled to the complementary transconductor: the residual output current from one stage was used to modulate the output node of the other, and the same arrangement was applied symmetrically in the reverse direction. This bidirectional cross-coupling was intended to exploit the complementary transconductance to extend the effective dynamic range of the conversion, allowing each stage to reinforce the other at input voltages near the boundary of its individual valid operating region.

Despite the functional merit of this scheme, its practical realisation proved incompatible with the target supply voltage of $V_{DD} = 0.8$ V. A standard five-transistor OTA requires a minimum headroom that accommodates the gate-to-source voltage drop of the differential pair, the saturation voltage of the tail current source, and the voltage drop across the active load. At 0.8 V, this budget is already severely constrained. The cross-coupled dynamic range extension mechanism required one additional transistor in the critical signal path beyond the five-transistor baseline, and this single additional device was sufficient to push the cumulative headroom demand beyond the available supply, preventing the circuit from maintaining all active elements in saturation over the intended input range and thus rendering the architecture non-functional in practice.

In an effort to recover the lost headroom margin, the conventional current mirror structures within the OTA branches were replaced by the low-voltage current mirror topology described by Razavi [**razavi**]. This configuration reduces the minimum voltage required across the mirror branch by eliminating one stacked gate-to-source drop from the critical headroom path, at the cost of a moderate increase in circuit complexity. However, even with this substitution, the cumulative voltage demand of the complete circuit encompassing the differential input pair, the tail current source, the low-voltage mirror, and the cross-coupled output branch remained in excess of 0.8 V. No viable bias point could be identified that simultaneously maintained all transistors in their correct operating region, and the architecture was consequently abandoned.

3.4 Phase III: The Final Adopted Architecture

3.4.1 Five-Transistor OTA Core and Optimized Transistor Sizing

3.4.2 Ratioed Output Slicers Design

3.4.3 Asynchronous Control Logic and Clockless Event Generation Network

Following the rejection of the complementary dual-input approach, the design converged on a classical five-transistor operational transconductance amplifier as the foundation for the comparator. In this topology, the differential pair is loaded by an active current mirror formed by two PMOS transistors: one connected in diode configuration, with its gate tied directly to its own drain, and a second transistor whose gate is driven by that same node. This arrangement constitutes the standard active load of a differential pair and is the defining structural element of the five-transistor OTA [**razavi**]. By dispensing with the additional stacked elements that were required by the previous candidate, this topology fits within the available headroom at $V_{\mathrm{DD}} = 0.8$ V and operates correctly across the intended input range.

To maximise the sensitivity of the comparator to small input voltage differences, the OTA was initially evaluated in open-loop configuration, where a high voltage gain could be obtained through appropriate transistor sizing. This open-loop approach, however, introduced a significant practical difficulty: even a small unintended differential voltage at the input—arising from device offset, thermal noise, or a marginal overdrive condition—was sufficient to drive the single-ended output rapidly into hard saturation before the circuit could resolve the intended comparison. Once the output reached the supply rail or ground, the recovery time was limited by the bias current available to discharge or charge the high-impedance output node, and this prolonged saturation state degraded both the effective comparison speed and the consistency of successive conversions.

To resolve this behaviour, the OTA was reconfigured as a unity-gain voltage follower by connecting the output directly to the inverting input, forming a 100% negative feedback loop. In this configuration, the feedback action drives the differential input of the OTA to near zero in steady state, which constrains the output to track V_{in} closely and keeps the amplifier permanently within its linear operating region, thereby preventing the saturation that had affected the open-loop scheme. The voltage follower thus acts as a low-impedance replica of the input signal, reproducing V_{in} at its output with the accuracy determined by the open-loop gain and the input-referred offset of the OTA.

The actual digitisation of V_{in} is delegated to a bank of CMOS slicer inverters, each driven directly by the follower output. Rather than relying on asymmetric current mirror ratios to encode the decision levels—as in the CMF-ADC—the present architecture encodes each comparison threshold through the PMOS-to-NMOS width ratio of the corresponding inverter, which determines its switching trip point V_M. For a CMOS inverter in which the relative transistor strengths are set by the ratio $r = (W/L)_p\,/\,(W/L)_n$, the trip point is given approximately by

$$V_M \approx \frac{V_{\text{DD}}\sqrt{r}}{1+\sqrt{r}}, \tag{3.1}$$

so that an inverter with a larger r exhibits a higher trip point, while a smaller r shifts V_M below $V_{\text{DD}}/2$. By assigning a distinct ratio to each output bit, the trip points of the slicer bank are distributed across the voltage axis, and the thermometer-coded output word is formed by the set of inverters whose input—the follower output—has or has not yet reached their respective thresholds.

In the implemented five-bit configuration, the most significant output bit, out_4, is formed by an inverter with a PMOS-to-NMOS ratio of $r = 1.5$, yielding a trip point above $V_{\text{DD}}/2 = 0.4$ V. Conversely, the least significant bit, out_0, uses a ratio of $r = 0.5$, placing its trip point below 0.4 V. The three intermediate bits are assigned ratios distributed between these extremes, so that the five trip points bracket the reference level of 0.4 V symmetrically. The operating principle is directly analogous to that of the CMF-ADC described in the first candidate: as V_{in} increases from zero toward V_{DD}, each slicer transitions from logic high to logic low in sequence, beginning with the inverter whose trip point is lowest—that is, the one whose threshold is crossed first—and ending with the inverter whose higher ratio causes it to hold its output high until a larger input voltage is reached. At $V_{\text{in}} = 0.4$ V, the slicers with $r > 1$ still output logic high while those with $r < 1$ have already transitioned to logic low, and the central slicer at $r = 1$ sits precisely at its trip point in a metastable condition. The resulting thermometer code at the reference level is consequently 11?00, in complete correspondence with the code produced by the CMF-ADC under the same input condition.

The principal advantage of this approach over the current-mirror architecture is its decoupling of the input buffering function from the threshold-setting function. The voltage follower provides a clean, low-impedance replica of V_{in} to all slicers simultaneously, while the trip-point distribution is controlled entirely through transistor sizing, which is a well-defined, lithographically precise parameter. This separation also simplifies the headroom analysis: the five-transistor OTA operates under unity-gain feedback at a common-mode level close to V_{in}, with no additional stacking requirements, while the slicer inverters are single-stage structures that impose no additional headroom constraint beyond the supply rails.

3.5 Analytical Modeling and Design Constraints

3.5.1 Maximum Input Wave Frequency and Slope Overload

3.5.2 Propagation Delay and Slicer Response Time Modeling

3.5.3 Resolution, Power Dissipation, and Voltage Headroom Trade-offs

3.6 Chapter Summary

Chapter 4

Synchronous Reference Flash ADC

4.1 Motivation for Architectural Selection

Benchmarking a novel converter architecture requires a reference point that is close enough structurally to make the comparison genuinely informative. The synchronous reference used in this work is a 4-bit Flash ADC with rail-to-rail comparators, originally presented in [**flashadc_ref**] and designed in a 65-nm CMOS process. For the purpose of this comparison, the architecture was re-implemented and simulated in a 16-nm technology node, under the same conditions applied to the proposed asynchronous converter throughout this dissertation.

The choice of this particular design was not arbitrary. Several properties directly align with those of the asynchronous architecture described in Chapter **??**, and it is precisely this alignment that makes the comparison meaningful. Both converters use the Flash topology, so the number of comparators and the fundamental conversion principle are shared. Both are designed for 4-bit resolution, keeping the complexity of the comparison logic identical. Both feature rail-to-rail input comparators, ensuring that the input voltage swing does not introduce any asymmetry between the two designs. Finally, both architectures are explicitly oriented towards low-power operation, which is the design space where the asynchronous approach is expected to have the most significant impact.

The fundamental difference, and the one this comparison is intended to illuminate, is the presence of a global clock. The synchronous reference depends on a periodic clock to coordinate its comparators and define valid output instants, while the asynchronous architecture eliminates the clock entirely. Removing the clock distribution network is one of the primary motivations for adopting asynchronous operation, since the clock tree contributes a non-trivial fraction of total power in high-speed designs. By holding all other parameters constant and changing only this aspect, the comparison provides a direct and controlled view of what asynchronous operation actually costs or saves in power and performance.

4.2 Architecture Overview

A Flash ADC resolves an analogue input voltage in a single conversion step by comparing it simultaneously against $2^N - 1$ evenly spaced reference levels, where N is the number of output bits. For a 4-bit converter, this means 15 comparators operate in parallel at every active clock edge, each producing a binary decision about whether the input exceeds its assigned threshold. The outputs of all 15 comparators form a thermometer code a sequence of ones followed by zeros with no transitions other than the boundary between them which is then passed to an encoder that maps it to the final 4-bit binary word.

[Insert: top-level block diagram of the synchronous Flash ADC]

Figure 4.1: Top-level block diagram of the synchronous 4-bit Flash ADC [**flashadc_ref**].

The reference voltages fed to each comparator are generated by a resistor ladder connected between V_{REF+} and V_{REF-}, which in this rail-to-rail implementation are tied directly to the supply rails V_{DD} and V_{SS}. The ladder divides the full input range into 15 equally spaced steps of one LSB each, so the k-th comparator from the bottom receives a reference of

$$V_{REF,k} = V_{SS} + k \cdot \frac{V_{DD} - V_{SS}}{2^N}, \qquad k = 1, 2, \ldots, 2^N - 1. \tag{4.1}$$

Since all comparators are driven by the same clock, every decision is made within a single clock period and the output is valid only at well-defined time instants. This is the defining characteristic of a synchronous converter and stands in direct contrast to the asynchronous operation discussed in Chapter **??**.

4.3 The StrongARM Latch Comparator

4.3.1 Circuit Description and Operating Principle

The comparator used in this architecture is the StrongARM latch, a dynamic comparator topology that has become something of a standard reference in both high-speed receiver design and data converter circuits [**razavi_strongarm**]. Its widespread adoption reflects a genuinely attractive combination of characteristics: high regeneration speed, good noise performance, and essentially zero static power dissipation all of which matter in a Flash ADC where 15 comparators must operate in parallel.

[Insert: StrongARM latch schematic]

Figure 4.2: Schematic of the StrongARM latch comparator [**flashadc_ref**].

The circuit operates in two alternating phases controlled by the clock signal a reset phase and an evaluation phase.

The reset phase occurs when the clock is low, the tail transistor is turned off and a pair of reset switches pull both output nodes up to V_{DD}. This precharges the outputs to a known state before every comparison, so that each evaluation begins from the same initial condition regardless of the previous result. Because the tail transistor is off and no current path exists through the core of the circuit, static power dissipation is identically zero during this phase.

The evaluation phase occurs when the clock rises, the tail transistor turns on and current begins to flow through the differential input pair. The input transistors steer this current according to the voltage difference $\Delta V = V_{INP} - V_{INN}$: the side with the higher input voltage conducts more, causing that output node to discharge faster than the other. Once a sufficient imbalance develops across the internal nodes, the cross-coupled inverter pair at the output enters regeneration a positive feedback mechanism that amplifies the initial imbalance exponentially until the outputs settle at valid logic levels [**razavi_strongarm**]. The time needed to complete the regeneration is inversely related to the input overdrive: larger differences between the input voltage and the reference threshold resolve faster, while inputs very close to the threshold require more time. This input-dependent latency is a fundamental property of dynamic comparators and is one of the key differences exploited by the asynchronous architecture discussed later.

Since current only flows during the evaluation phase, and only briefly at that, the average power consumption is directly proportional to the clock frequency. This makes the StrongARM latch well suited to low-power converter design: reducing the sampling rate reduces the power linearly.

4.3.2 Rail-to-Rail Input Range

A key property of this comparator implementation is its ability to correctly resolve input voltages across the full supply range, from V_{SS} to V_{DD}. In a Flash ADC, the comparators assigned to the top and bottom reference levels must operate very close to the supply rails, and a comparator that loses sensitivity or gain in those regions would introduce nonlinearity at the extremes of the transfer characteristic, degrading the integral linearity of the converter. The rail-to-rail design ensures this does not happen, maintaining consistent performance across the entire input range.

This property is shared by the asynchronous architecture presented in this work, so it does not introduce any asymmetry into the comparison between the two designs.

4.4 Clock Generation

Since the architecture is synchronous, a stable clock signal must drive the comparator array and determine the conversion instants. For simulation purposes, the clock was generated using a ring oscillator, which provides a compact and realistic on-chip clock source without requiring an ideal external generator.

Rather than driving the comparators directly from the oscillator output, the signal was routed through a tapered inverter buffer chain designed to model the loading and parasitics of a realistic clock distribution network. The chain consists of a sequence of inverters with progressively increasing drive strength, with each stage sized at approximately twice the fanout of the previous one. The final stage of the chain is a $\times 24$ fanout inverter, whose size reflects the capacitive load that a clock signal would encounter when distributed through the metal routing to the ADC core the interconnect wires, vias, and comparator gate inputs that the clock must charge and discharge on every cycle.

A $\times 24$ final buffer is a reasonable and representative choice for modelling the routing from a clock source to the ADC, as it captures the kind of loading one would expect in a physically compact but realistic clock distribution. Neglecting this stage would result in a power estimate that is overly optimistic: in a real implementation, the clock tree is rarely a negligible contributor to total power, and any fair assessment of the synchronous architecture must account for it.

The complete clock path used in simulation is therefore: ring oscillator buffer chain ($\times 2$ fanout per stage) $\times 24$ output buffer comparator array.

4.5 Power Measurement Methodology

To ensure the fairest possible comparison with the proposed asynchronous architecture, the power figures reported in this chapter cover only the core analogue conversion path. Specifically, the encoder the digital logic that translates the 15-bit thermometer code output of the comparators into the final 4-bit binary word was excluded from all power measurements.

The reasoning is straightforward, the asynchronous architecture presented in this work also does not include encoder power in its reported figures. Including it on one side of the comparison but not the other would artificially inflate the synchronous power budget and lead to a conclusion that does not fairly reflect the underlying architectural trade-off. Since both designs require some form of thermometer-to-binary conversion at their outputs, and since this encoding logic is largely independent of whether the preceding conversion is synchronous or asynchronous, the most defensible approach is to exclude it from both and compare only the converter cores.

The power reported in Table 4.1 therefore includes the resistor ladder, the 15 StrongARM latch comparators, the ring oscillator, and the tapered clock buffer chain described in the previous section.

4.6 Simulation Results

Table 4.1: Simulated performance summary of the synchronous reference Flash ADC in 16-nm CMOS.

Parameter	**Value**	**Unit**
Supply Voltage		V
Resolution		bit
Sampling Frequency		MS/s
SNDR		dB
ENOB		bit
Power Consumption		μW
Rail-to-Rail Input		Yes
FoM$_W$ (Walden)		fJ/conv.-step

Chapter 5

Trimming

5.1 Body Biasing Trimming

5.1.1 Function of NMOS and PMOS Transistors Under Different Body Bias

The threshold voltage of a MOSFET is not solely determined by the device geometry and doping profile but is also a function of the potential applied to its body terminal relative to the source. This dependence, commonly referred to as the body effect or substrate bias effect, is described for an NMOS transistor by

$$V_{tn} = V_{tn0} + \gamma_n \left(\sqrt{2\phi_{F,n} + V_{SB}} - \sqrt{2\phi_{F,n}} \right), \tag{5.1}$$

where V_{tn0} is the zero-bias threshold voltage, γ_n is the body effect coefficient, $\phi_{F,n}$ is the Fermi potential, and V_{SB} is the source-to-body voltage. The analogous expression for a PMOS transistor, accounting for the sign conventions appropriate to a p-channel device, takes the form

$$|V_{tp}| = |V_{tp0}| + \gamma_p \left(\sqrt{2\phi_{F,p} + |V_{SB}|} - \sqrt{2\phi_{F,p}} \right), \tag{5.2}$$

where the relevant source-to-body voltage for the PMOS is determined by the potential difference between the source and the N-well. In both cases, an increase in $|V_{SB}|$ raises the magnitude of the threshold voltage, a condition referred to as reverse body biasing (RBB); conversely, reducing $|V_{SB}|$ below its nominal value or, in technologies that permit it, forward-biasing the body junction lowers $|V_t|$, a condition known as forward body biasing (FBB).

A direct consequence for the slicer inverters in the proposed architecture is that each cell's trip point, given by equation (3.1), depends on the threshold voltages of both transistors composing the inverter. By substituting the body-bias-dependent expressions for V_{tn} and $|V_{tp}|$ into the inverter threshold equation and expanding to first order, it is apparent that a positive shift in V_{tn} achieved by applying RBB to the NMOS weakens the pull-down network relative to the pull-up network and therefore raises V_M, while an equivalent increase in $|V_{tp}|$ weakens the pull-up network and lowers V_M. This asymmetry between the NMOS and PMOS body bias responses provides two independent degrees of freedom for adjusting the trip point of each slicer: the NMOS bulk voltage controls the effective strength of the pull-down path, while the PMOS bulk voltage (equivalently, the N-well potential) controls the pull-up path. Applying corrections simultaneously to both body terminals therefore allows independent tuning of the trip point magnitude and of the symmetry of the transition characteristic, providing a richer correction space than either terminal alone could offer.

5.1.2 Process Corner Dependence and Motivation for Trimming

In a deep-submicron CMOS process, fabrication variability causes the threshold voltages and carrier mobilities of the NMOS and PMOS devices to deviate from their nominal (typical-typical, TT) values. The standard process corners used to bound this variability slow-slow (SS), fast-fast (FF), fast-NMOS slow-PMOS (FS), and slow-NMOS fast-PMOS (SF) capture the four extreme combinations of NMOS and PMOS behaviour and represent the conditions under which circuit performance is most likely to degrade.

For the slicer-based thermometer coder described in the previous chapter, the impact of process variation is particularly significant because the thermometer code relies on the precise relative ordering of the inverter trip points across the output bit bank. At the typical corner, the ratios r_k assigned to each slicer place the trip points at the intended positions relative to the reference level of 0.4 V, and the resulting differential non-linearity (DNL) and integral non-linearity (INL) of the converter remain within specification. At non-typical corners, however, the effective transistor strengths deviate from their nominal values in a manner that is not uniform across bits: since the PMOS and NMOS devices within each slicer are affected differently depending on the corner, the trip point of each inverter shifts by a bit-dependent amount, breaking the intended monotonic spacing and introducing non-linearities that manifest as elevated DNL and INL.

At the SS corner, both NMOS and PMOS threshold voltages increase, reducing the drive strength of both pull-down and pull-up networks. Although the symmetric nature of this degradation might suggest a self-compensating effect on the trip point, the body effect coefficients γ_n and γ_p and the nominal threshold values are in general not matched between the two device types, so the net shift in V_M is non-zero and, more importantly, bit-dependent due to the different sizing ratios across the slicer bank. The FF corner produces the complementary effect: reduced threshold voltages increase the drive strength of both device types, again shifting the trip points by amounts that vary across the bit cells. The FS and SF corners are qualitatively different in that they introduce an asymmetric perturbation: at FS, the NMOS is faster than nominal while the PMOS is slower, biasing all slicers toward a lower trip point and compressing the effective decision levels toward the bottom of the input range; at SF, the complementary imbalance expands the decision levels toward the top. In all four non-typical corners, the degradation in linearity was found to cause DNL excursions in excess of 0.5 LSB, rendering the converter non-monotonic under these conditions.

5.1.3 Offline Trimming Strategy via Body Bias Adjustment

To restore the linearity of the thermometer coder across process corners without modifying the core circuit topology, an offline trimming strategy based on body bias adjustment was adopted. The principle of the approach is to apply distinct bulk voltages to the NMOS and PMOS transistors of the output slicer stage independently for each device type in a manner calibrated to shift the trip points back toward their nominal positions. Since both the NMOS bulk voltage V_{BN} and the PMOS bulk voltage V_{BP} contribute independently to the trip point through equations (5.1) and (5.2), respectively, adjusting both simultaneously allows correction of both the common-mode shift in trip point which affects DNL and the differential imbalance between NMOS-dominated and PMOS-dominated corners which affects INL. The trimming is defined as offline in the sense that the bulk voltages are determined once, based on the identified process corner, and held constant during normal operation; no continuous adaptive feedback is involved.

For each of the four non-typical corners, a specific pair of values (V_{BN}, V_{BP}) was determined such that the DNL of the converted thermometer code falls below 0.5 LSB across the full input range. At the SS corner, where both device types exhibit elevated threshold voltages and therefore reduced drive strength, forward body biasing is applied to both the NMOS and PMOS, reducing V_{tn} and $|V_{tp}|$ toward their nominal values and thereby restoring the intended trip point positions. At the FF corner, reverse body biasing is applied to both types to counteract the excessive drive strength resulting from the low threshold voltages. The FS and SF corners require asymmetric corrections: at FS, a stronger reverse bias is applied to the NMOS bulk, weakening the pull-down network to compensate for the fast-NMOS tendency, while the PMOS bulk is forward-biased to partially recover the reduced pull-up strength; the SF corner is handled symmetrically, with the dominant correction applied to the PMOS bulk.

The trimming was implemented in simulation using a Verilog-A behavioural model. The model monitors the process corner identifier and, based on the detected corner, drives V_{BN} and V_{BP} to the pre-computed correction values. This approach allows the trimming to be evaluated across all corners within the same simulation framework as the nominal design, without requiring modifications to the transistor-level netlist. The Verilog-A abstraction also provides a clean interface for extending the trimming table to intermediate corner points or to temperature-corner combinations should a finer correction granularity be required in future iterations. The effect of the body bias corrections on the DNL and INL of the converter is presented in Chapter **??**.

Chapter 6

Simulation Results and Performance Benchmarking

6.1 Simulation Setup and Input Test Signals

6.2 Static Linearity Results

6.3 Power Dissipation and Energy Efficiency

6.4 Comparative Analysis and Discussion

Chapter 7

Conclusions and Suggestions for Future Work

7.1 Summary of the work developed

7.2 Concluded Objetctives

7.3 Future Work

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